Sri Lanka Institute of Information Technology

Final Project Report
Web-based Laundry Management System

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Group Members

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1. Introduction

Project Overview

The Laundry Management System is a web-based application designed to automate and simplify a laundry business's daily operations. Traditional laundry services mainly rely on manual procedures like handwritten orders, paper receipts, and in-person interactions between employees and clients. Delays, lost orders, miscommunication, and overall inefficiency are frequently the results of manual operations.

By developing an integrated platform that allows users to easily place orders online, arrange pickups and deliveries, make secure payments, and monitor the status of their laundry, our system aims to address these issues. On the other hand, a centralized dashboard allows the administrator to oversee orders, personnel, services, payments, and inventory.

By automating essential procedures and offering real-time updates, the system enhances operational efficiency, guarantees accurate record-keeping, and improves customer experience.

Objectives

- To develop an efficient and user-friendly platform for handling laundry orders.
- To automate essential procedures like staff task assignment, billing, and pickup/delivery scheduling.
- To improve the client experience by using digital payments, flexible scheduling, and order tracking.
- To provide the administrator with the resources they need to manage staff, services, inventory, and financial reports.
- To reduce human error and manual process delays.

Target Users

1. **Customer**
   Customers are the primary end-users who place laundry orders, schedule pickups, track their laundry status, make payments, and provide feedback. Their satisfaction is crucial to business success, making them the most important drivers of system usage.

2. **Manager**
   The laundry manager oversees daily operations, including service availability, order processing, and task assignments to staff. They can update service menus (e.g., wash, dry clean, ironing), manage workload distribution, and monitor order timelines. They also use reports to evaluate operational efficiency.

3. **Delivery Coordinator**
   Responsible for planning and assigning pickup and delivery routes. The delivery coordinator ensures that laundry is collected and delivered on time, manages driver schedules, and updates customers about their order status.

4. **Admin**
   The admin is the top-level controller of the system. They manage user accounts, define access levels, update service categories, set pricing, and monitor overall system performance. Admins also generate and review detailed reports on orders, payments, and staff productivity. They are responsible for ensuring system security, performing backups, and keeping the platform running smoothly.

5. **Laundry Staff Member**
   Laundry staff members handle the actual washing, drying, ironing, and packaging of clothes.
   And collecting laundry from customers at the scheduled time, delivering cleaned clothes. They receive orders through the system. Their responsibilities include ensuring quality cleaning, meeting turnaround times, and updating order status once processing is complete. They play a direct role in maintaining service quality and customer satisfaction.

6. **IT Support Assistant**

Maintains the technical health of the laundry management system. They resolve software issues, manage user access control, update security measures, and ensure the platform remains functional and up to date.

Scope

- Registering, placing orders, scheduling pickup and delivery times, paying, and viewing order history are all accessible to customers.
- The manager can view placed orders and assign tasks to staff.
- Employees can view their assigned tasks and update order statuses by logging in.
- The administrator can generate reports, manage all users, orders, services, pricing, and inventory.

Limitations

- **No SMS or email alerts**
  The system will not produce any SMS or email notifications to customers concerning the payments or tracking of laundry orders. All notices will be displayed on the dashboard of the web application.
- **Web-only access (no mobile app initially)**
  The system will only be accessible through the web. The website will be designed to be viewed on a laptop or desktop computer.
- **Payment gateway depends on third party integration**
  The system will only calculate and display the bill of the payment. The payments will be handled by third party platforms that are outside of the system.
- **Doesn't support online maps to locate customers**
  The system will only provide addresses and directions to customer residences. The laundry delivery team has to rely on third party applications such as Google maps to locate customers and deliver the laundry.
- **Limited Language Support in First Release**
  Since the system will initially support only English, non-English speakers will face difficulties using the system.

2. Requirements

Functional Requirements

- User Registration and Authentication
- Profile Management
- Role-Based Access
- Order Placement
- Order Tracking
- Order Modification/Cancellation
- Task assignment
- Pickup and Delivery Scheduling
- Payment Processing
- Generate Reports
- Business Analytics
- Customer Feedback management
- System Administration

Non-functional Requirements

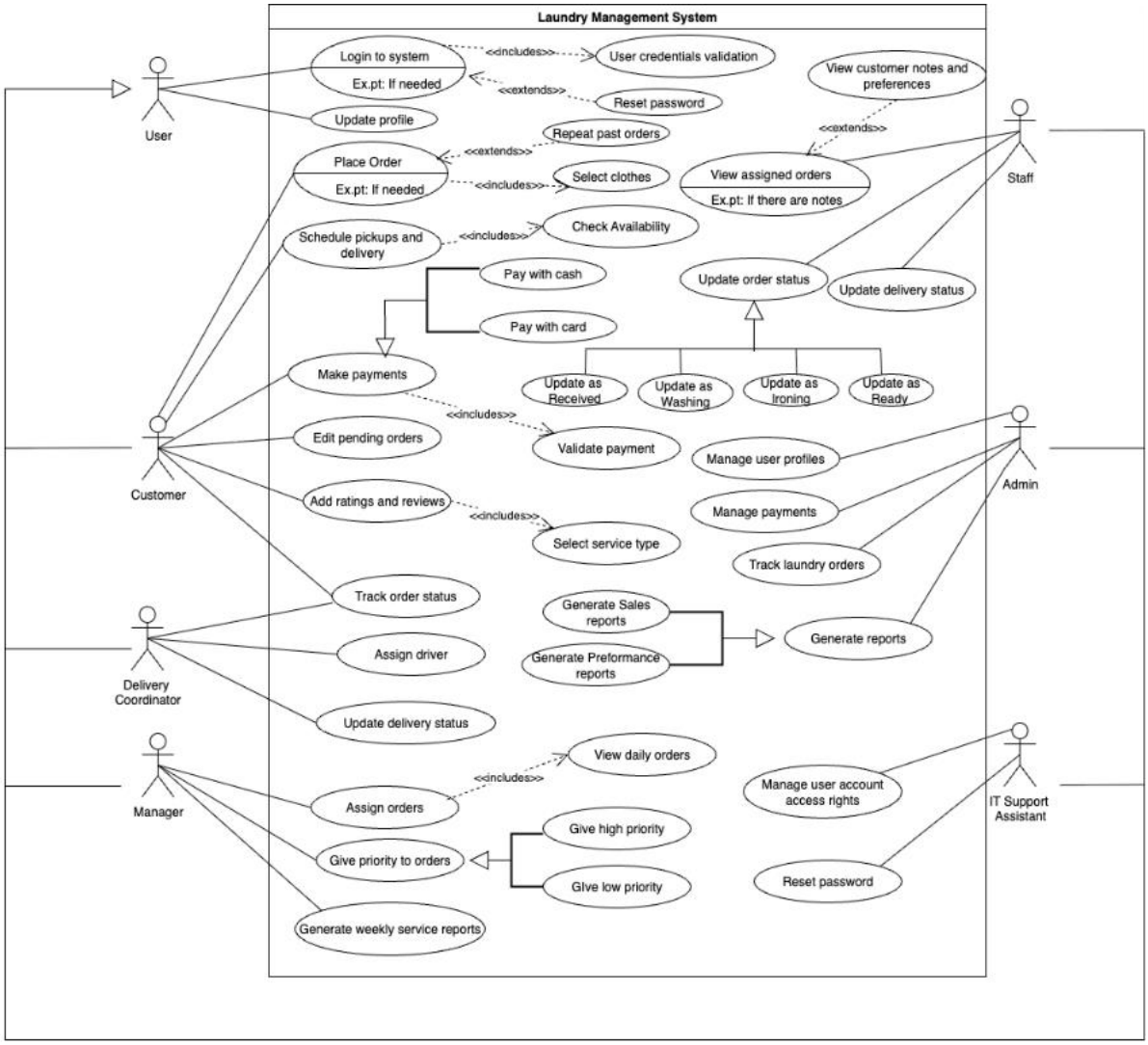
- **Performance**: The system will load fast, display accurate and up to date information. It will process laundry orders quickly to ensure a smooth customer experience.
- **Security**: The username and password will be asked from the user before logging into the system, providing security from unauthorized access.
- **Scalability**: The system will be able to handle the growing number of users so that it can be used in both small and medium-sized laundry.
- **Usability**: This web application will have a clean and responsive user interface, making it easy for the customer to use and interact with the system.
- **Reliability**: The system will have minimal downtime and stable hosting. If there's a system crash or error, the IT support assistants and admin will analyze the problem and fix the issue as quickly as possible, ensuring that users don't face any inconvenience.
- **Maintainability**: Modular, well-documented code and well planned, organized development processes will guarantee the maintainability of the system.

Constraints or Assumptions

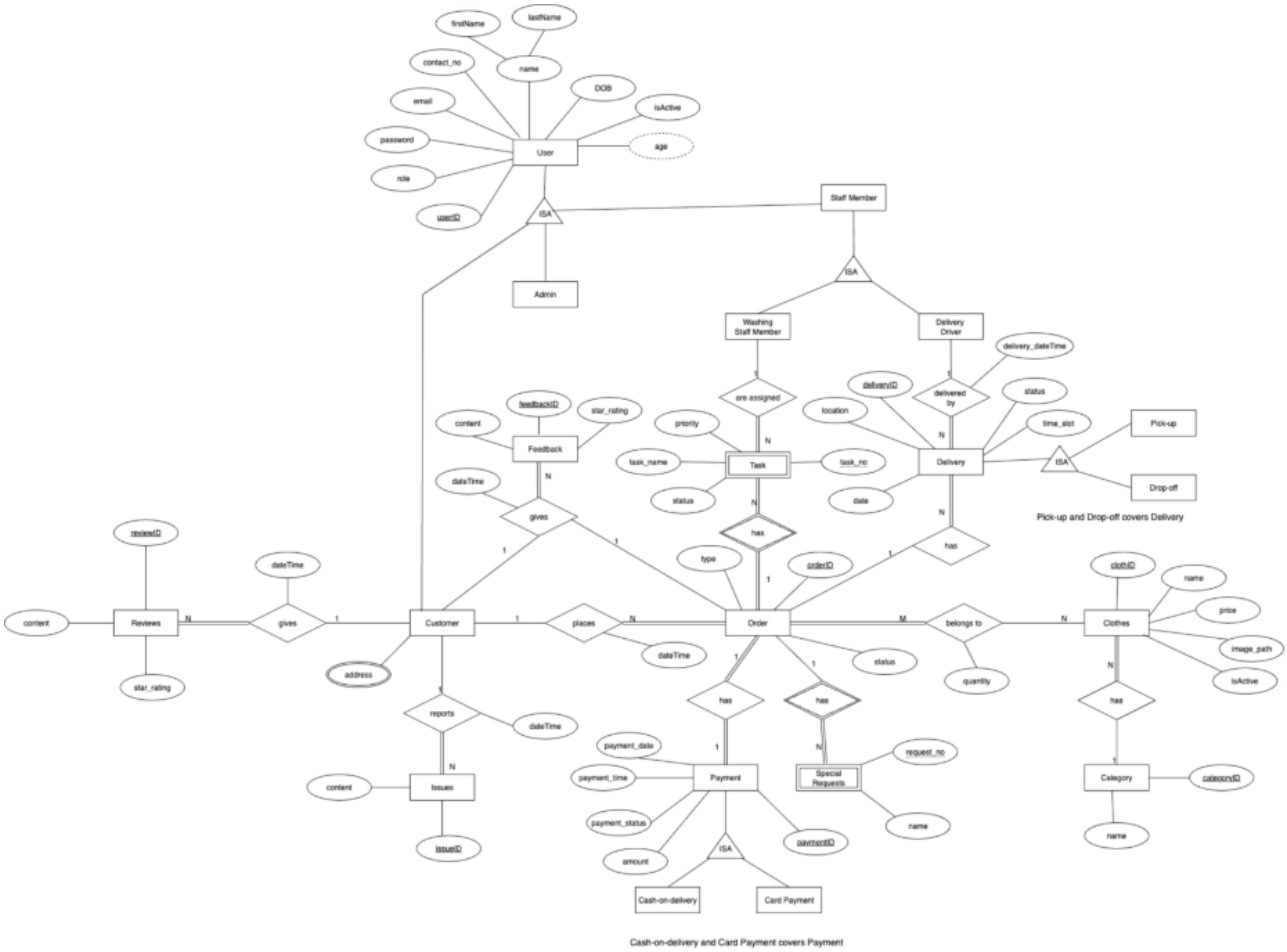
- Users have access to a modern browser.
- Internet connectivity is stable during order placement and payment.
- Customers provide accurate pickup/delivery addresses and contact numbers.
- Payment gateway is always available.
- All laundry machines are manually operated; no IoT integration.

3. Design

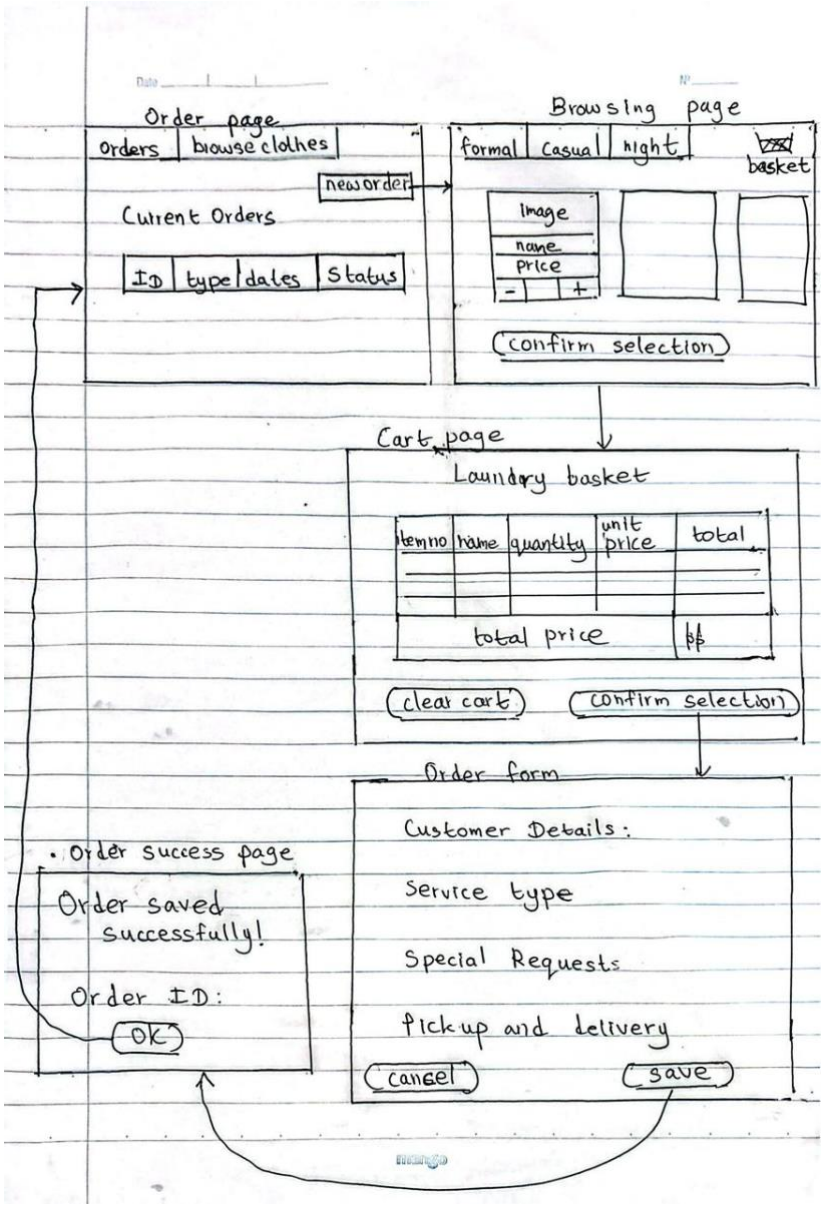
Use Case Diagram



ER diagram


firstName
lastName
contact_no
name
DOB
email
isActive
password
User
age
role
userID
ISA
Staff Member
Admin
ISA
Washing Staff Member
Delivery Driver
delivery_dateTime
are assigned
deliveryID
delivered by
status
feedbackID
star_rating
content
priority
location
time_slot
Pick-up
Feedback
Task
Delivery
ISA
task_name
task_no
Drop-off
dateTime
status
date
gives
has
has
Pick-up and Drop-off covers Delivery
reviewID
type
orderID
dateTime
clothID
name
content
Reviews
gives
Customer
places
Order
belongs to
Clothes
price
dateTime
image_path
address
status
quantity
isActive
star_rating
has
has
has
reports
dateTime
payment_date
request_no
content
Issues
payment_time
Payment
Special Requests
Category
categoryID
payment_status
name
paymentID
name
issueID
amount
ISA
Cash-on-delivery
Card Payment
Cash-on-delivery and Card Payment covers Payment

UI Sketches


Order page
Orders
browse clothes
neworder
Current Orders
ID
type
dates
Status
Browsing page
formal
casual
night
basket
image
name
price
-
+
confirm selection
Cart page
Laundry basket
itemno
name
quantity
unit price
total
total price
clear cart
confirm selection
Order form
Customer Details:
Service type
Special Requests
Pickup and delivery
cancel
save
Order success page
Order saved successfully!
Order ID:
OK

System Architecture

The system uses a layered Model-View-Controller (MVC) architecture to ensure modularity, scalability, and maintainability. It is divided into three main layers which are Business Logic Layer, Presentation Layer and Data Access Layer.

1. **Backend (Business Logic Layer)**

   In this layer, the code is structured into seven main packages including model, repository, service, controller, config, dto and handler, each of which handles a specific task in the system.

   i. **Model: c**ontains Java classes that represent entities in the system
   ii. **Repository:** provides an interface to interact with the database to perform CRUD operations
   iii. **Service:** contains the business logic of the application. It processes data from the repository, applies rules, and coordinates operations between components.
   iv. **Controller:** handles HTTP requests and responses, acting as the entry point for user interactions. It coordinates between the front end and the service layer.
   v. **Config:** defines application-wide settings, such as security configurations.
   vi. **Dto:** consists of objects used to transfer data between layers without exposing internal entity details.
   vii. **Handler:** processes incoming requests and coordinates responses.

2. **Frontend (Presentation Layer)**

   HTML, CSS, and JavaScript files are used to render the web pages of the system. They are each organized into separate directories for easy management. Thymeleaf framework is used to display dynamic data in the web application.

3. **Database (Data Access Layer)**

   The system uses a MySQL database to store data including user information, order and delivery details, and payment data. The Spring Data JPA framework and the JDBC template facilitate interaction between the Java application and the database.

4. Implementation

Tools and Technologies Used

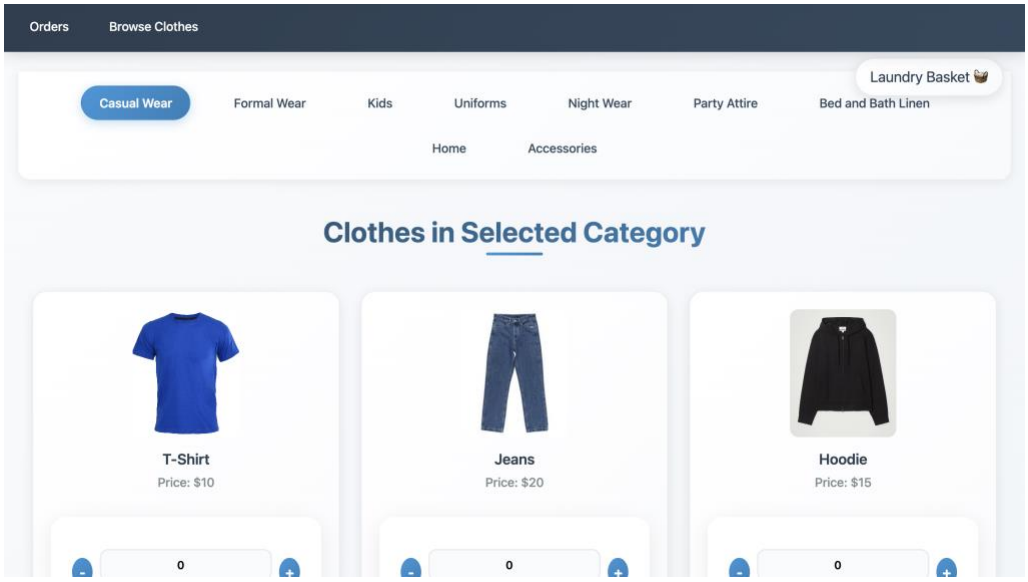
- **Programming languages:**
  - Backend- Java 17
  - Frontend- HTML, CSS, JavaScript, Thymeleaf
- **Frameworks:** Spring boot, Maven, Thymeleaf, Spring Data JPA, Spring Security, Spring Web
- **Server:** Apache Tomcat (embedded with Spring Boot)
- **Database:** MySQL
- **IDE:** Used IntelliJ IDEA Ultimate edition to develop the code
- **Deployment:** Deployed on a local server for development and testing
- **Version Control:** GitHub

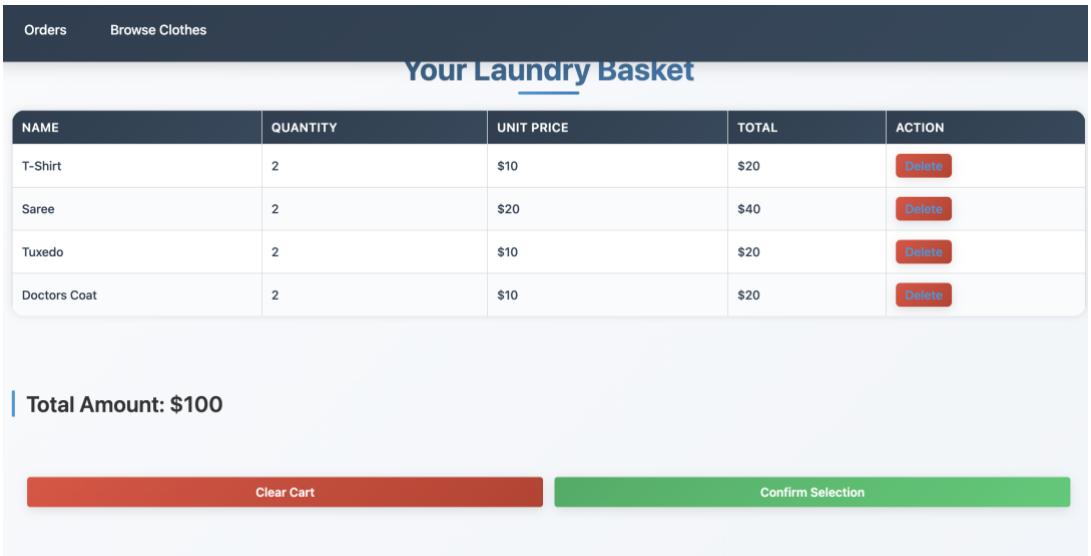
Key Features Developed

1. Laundry order management
2. User profile management
3. Pickup and delivery management
4. Payment management
5. Customer feedback and review management
6. Admin management

Screenshots of Core Functions

1) Order management





Orders Browse Clothes

Edit Order

Order ID: 16

Service Type:

washing

Pickup Date:

2025-10-25

Delivery Date:

2025-10-30

Update Order

Orders Browse Clothes

Create Laundry Order

Service Type

Washing

Special Requests

wash with care

Pickup and Delivery

Preferred Pickup Date:

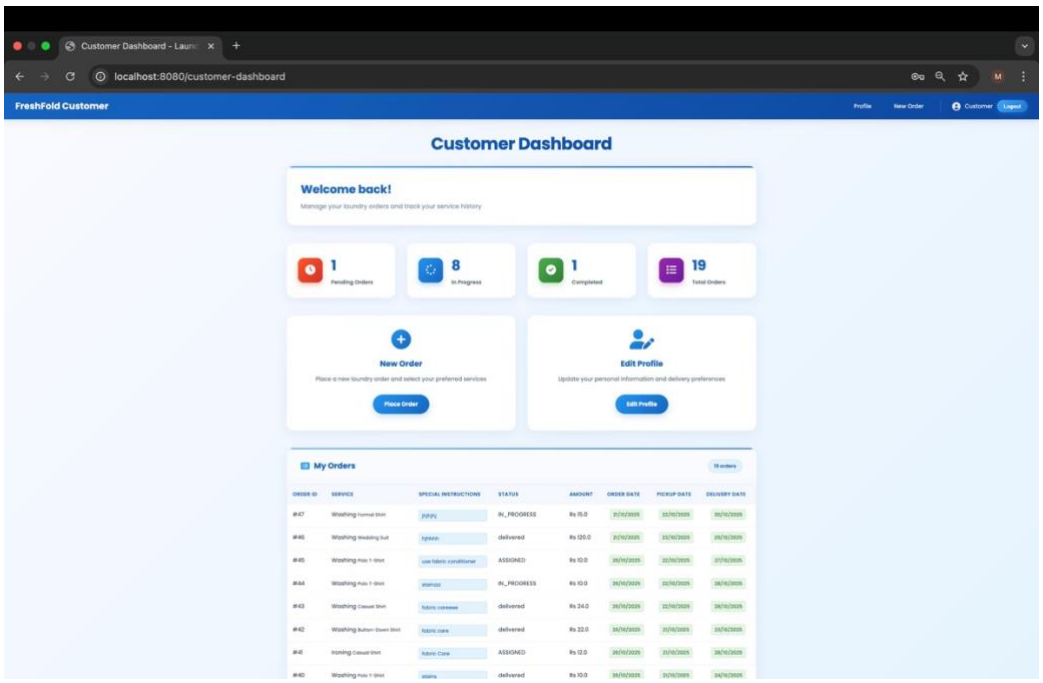
2025-10-28

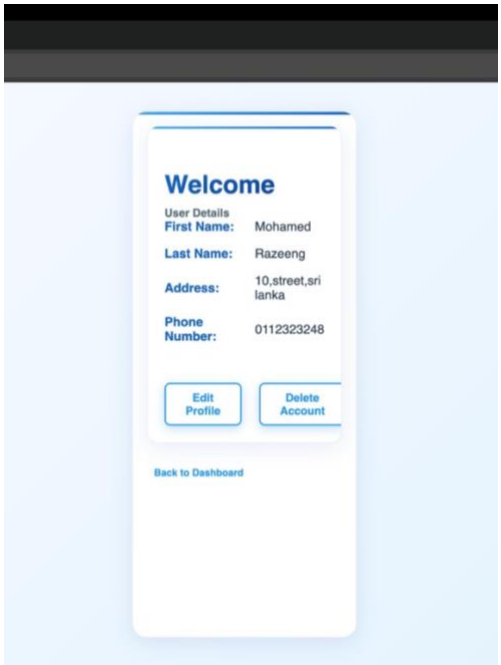
Orders Browse Clothes

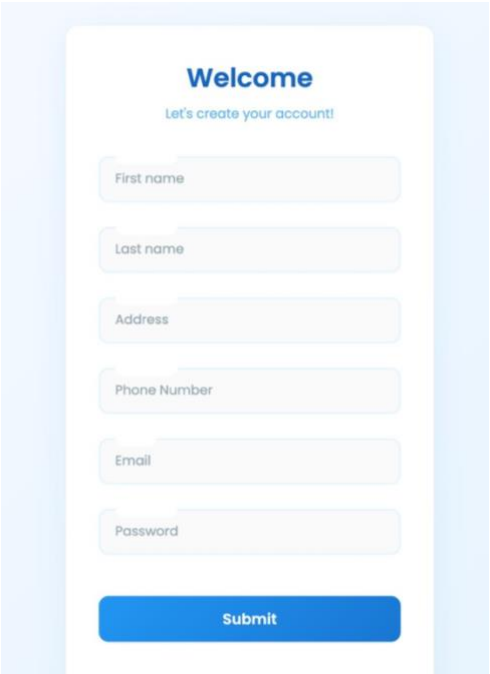
Selected Clothes

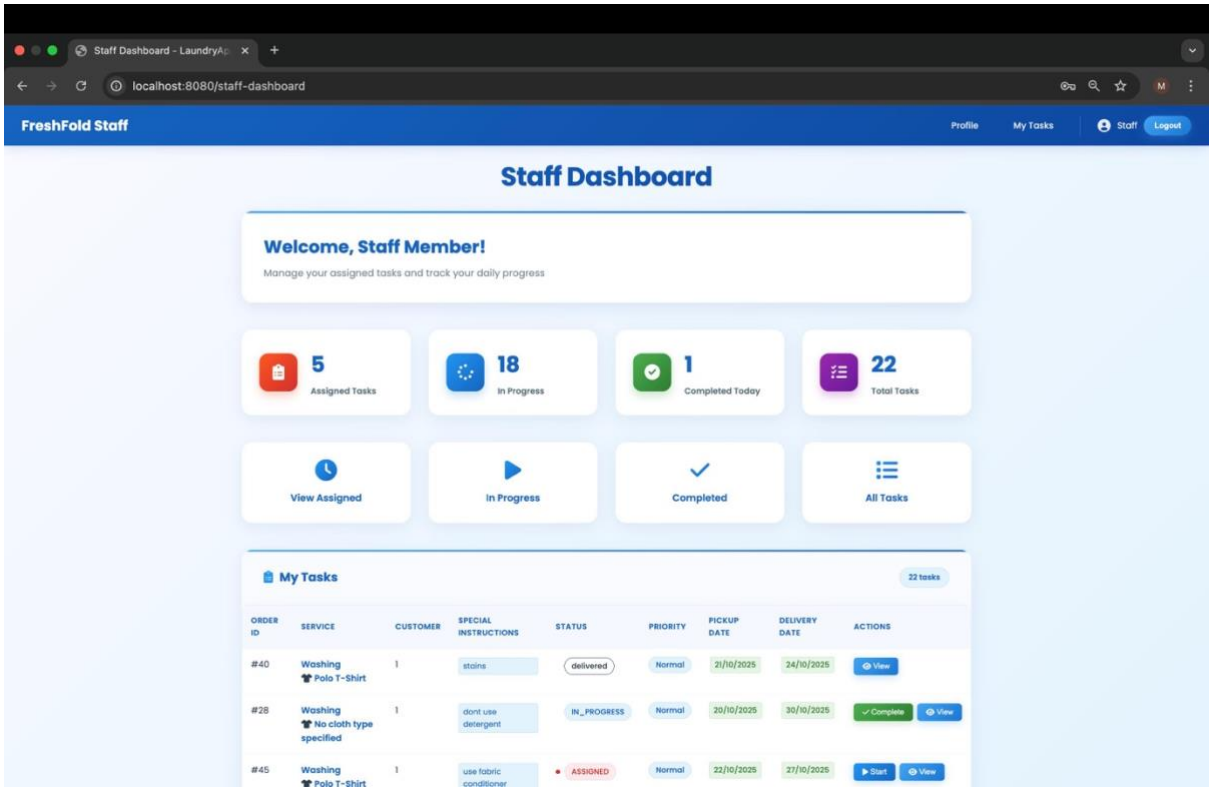
IMAGE	NAME	QUANTITY	UNIT PRICE	TOTAL	ACTION
	Suit	2 Update	$10	$20	
	Kids T-shirt	2 Update	$20	$40	

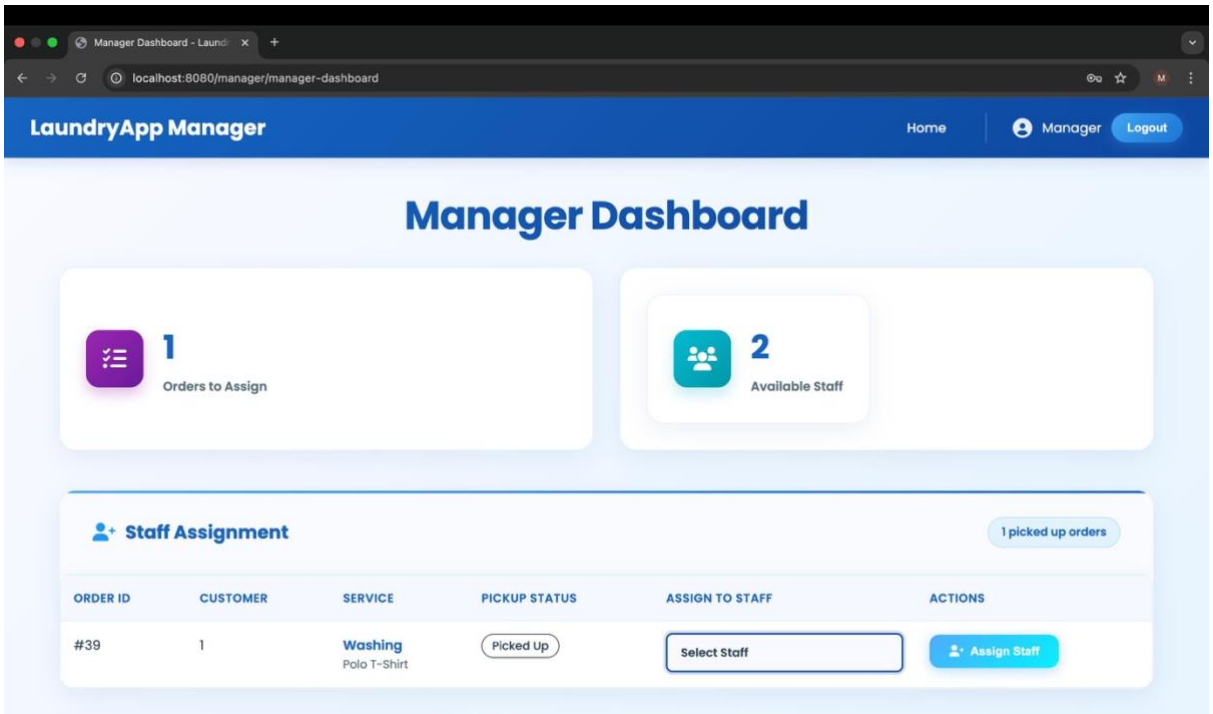
2) Profile Management



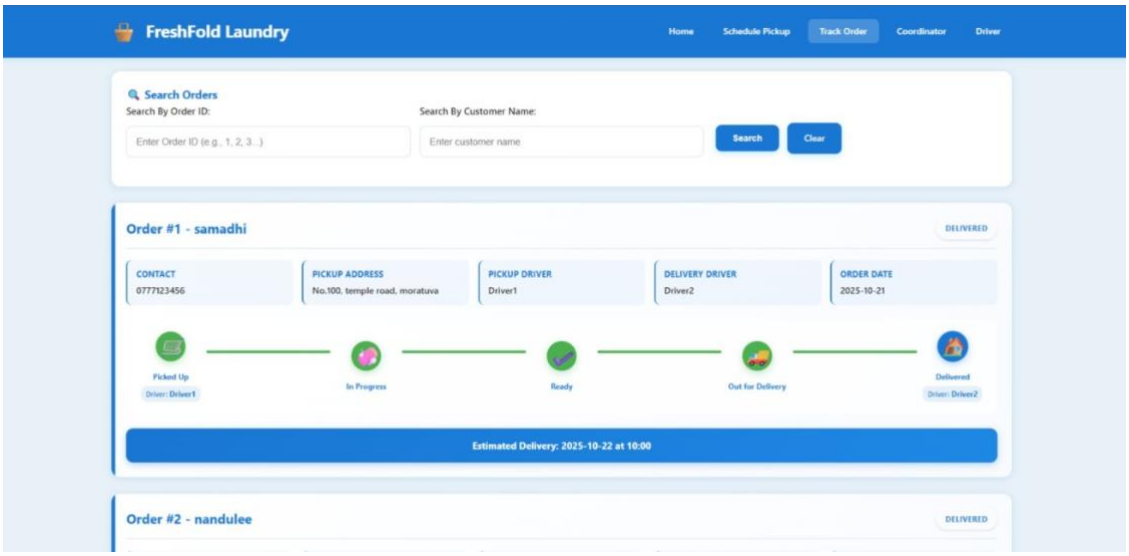


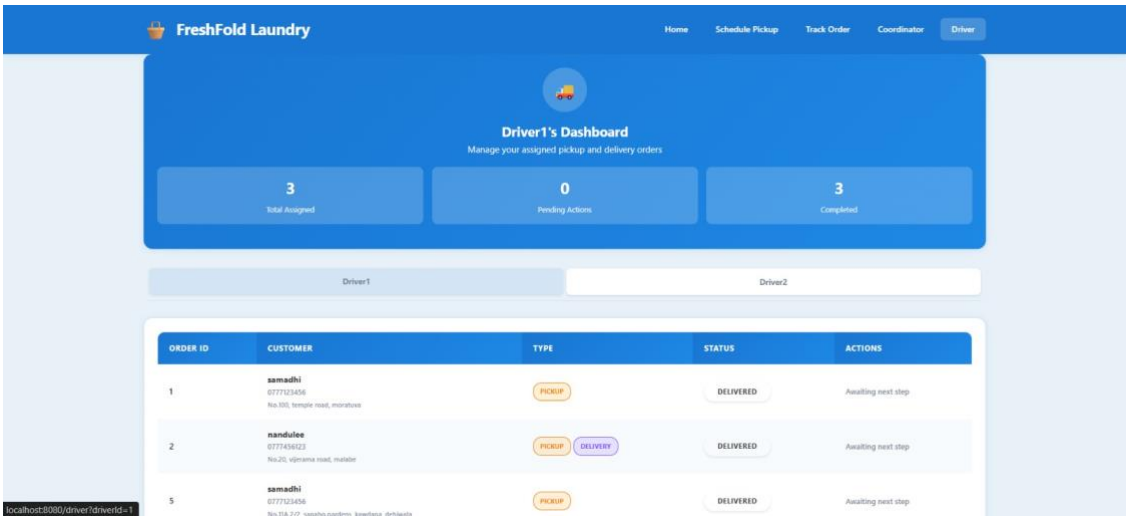


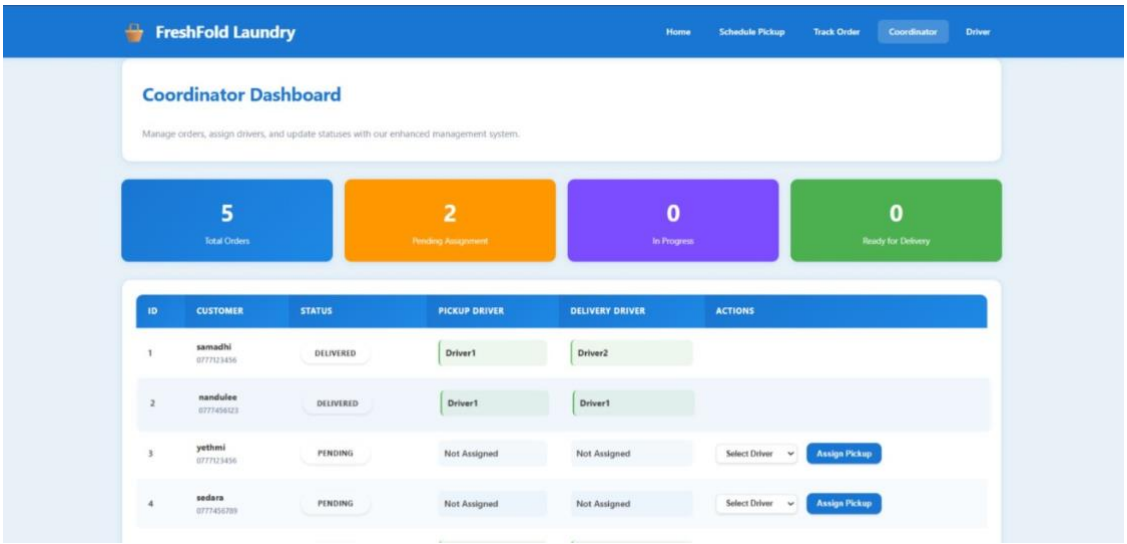

Staff Dashboard - LaundryAp
localhost:8080/staff-dashboard
FreshFold Staff
Profile
My Tasks
Staff
Logout
Staff Dashboard
Welcome, Staff Member!
Manage your assigned tasks and track your daily progress
5
Assigned Tasks
18
In Progress
1
Completed Today
22
Total Tasks
View Assigned
In Progress
Completed
All Tasks
My Tasks
22 tasks
ORDER ID
SERVICE
CUSTOMER
SPECIAL INSTRUCTIONS
STATUS
PRIORITY
PICKUP DATE
DELIVERY DATE
ACTIONS
#40
Washing
Polo T-Shirt
1
stains
delivered
Normal
21/10/2025
24/10/2025
View
#28
Washing
No cloth type specified
1
dont use detergent
IN_PROGRESS
Normal
20/10/2025
30/10/2025
Complete
View
#45
Washing
Polo T-Shirt
1
use fabric conditioner
ASSIGNED
Normal
22/10/2025
27/10/2025
Start
View

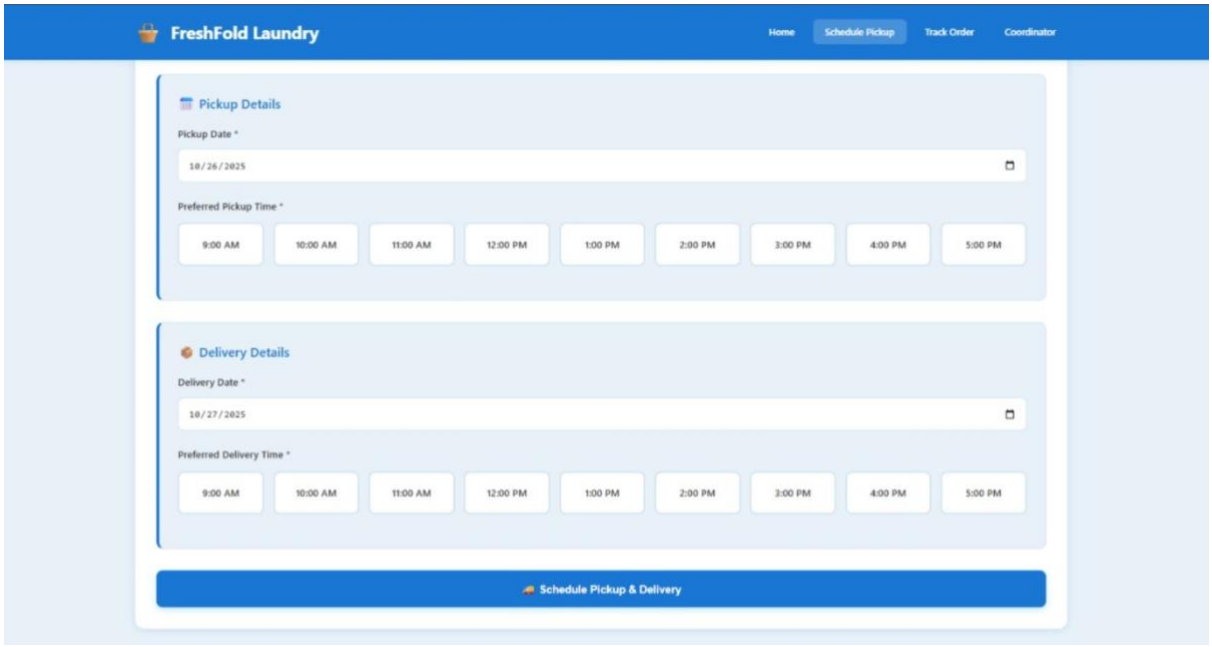

Manager Dashboard - Laund
localhost:8080/manager/manager-dashboard
LaundryApp Manager
Home
Manager
Logout
Manager Dashboard
1
Orders to Assign
2
Available Staff
Staff Assignment
1 picked up orders
ORDER ID
CUSTOMER
SERVICE
PICKUP STATUS
ASSIGN TO STAFF
ACTIONS
#39
1
Washing
Polo T-Shirt
Picked Up
Select Staff
Assign Staff

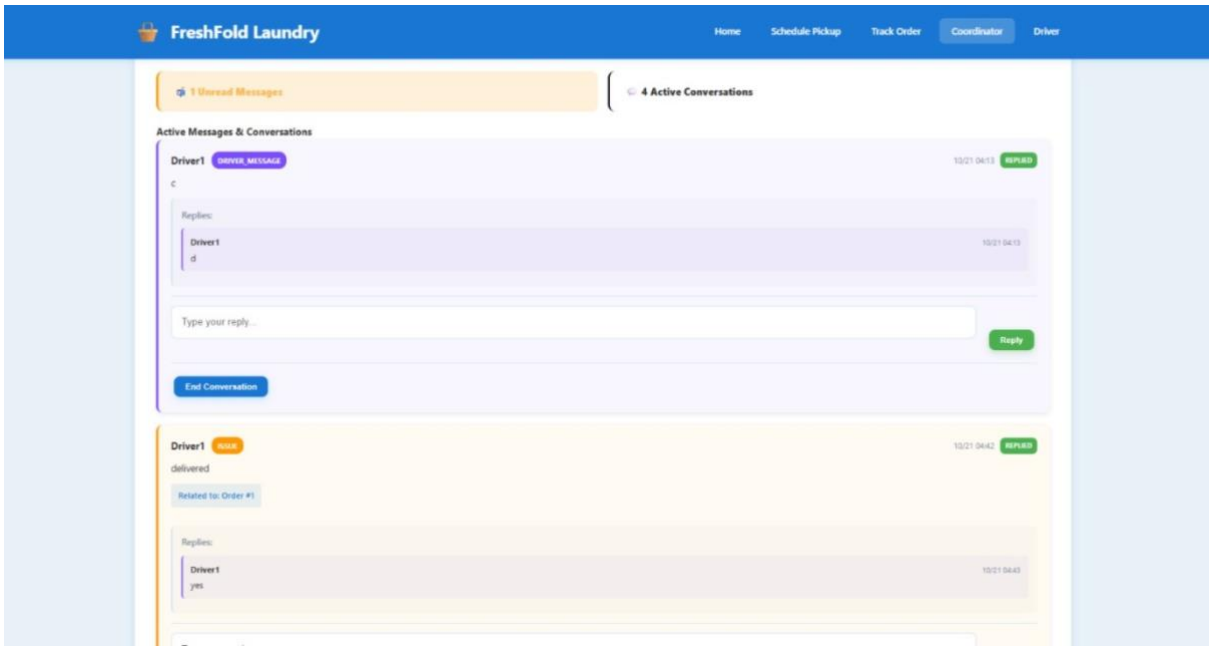
3) Pickup and delivery management



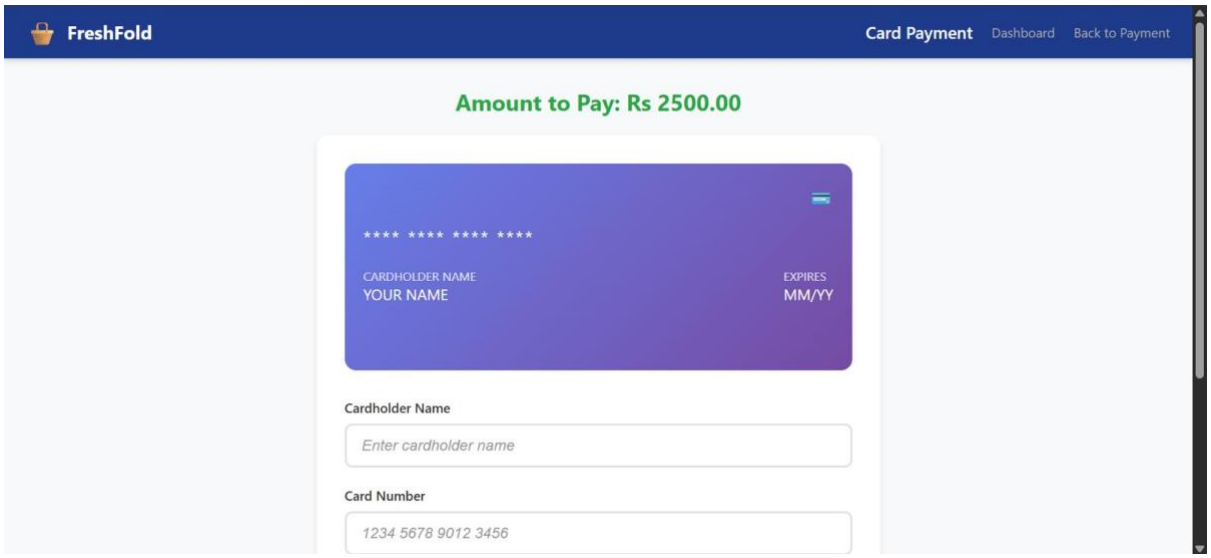


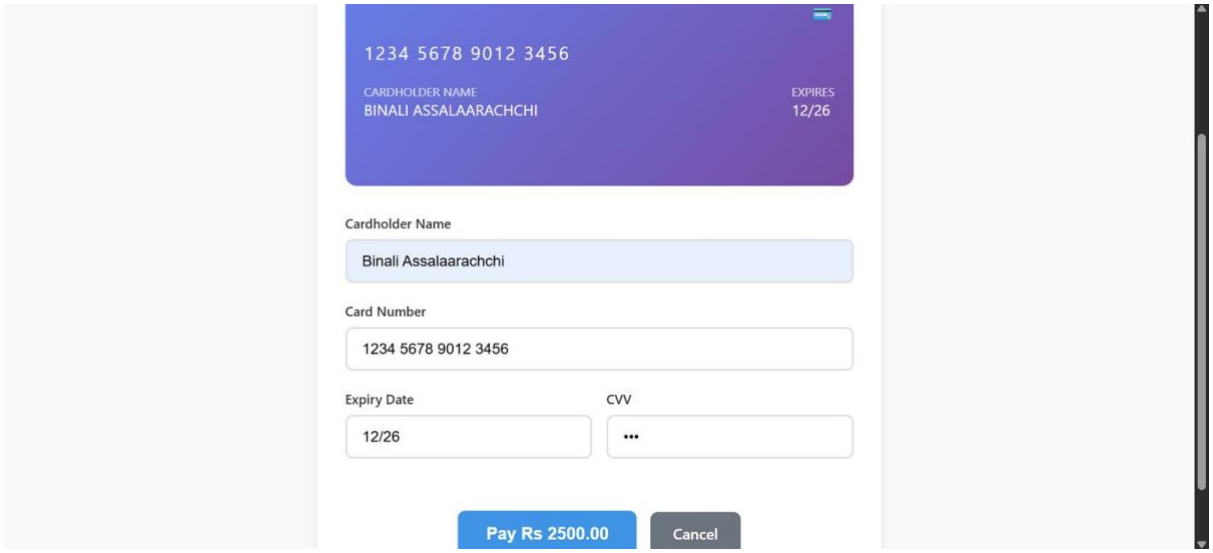


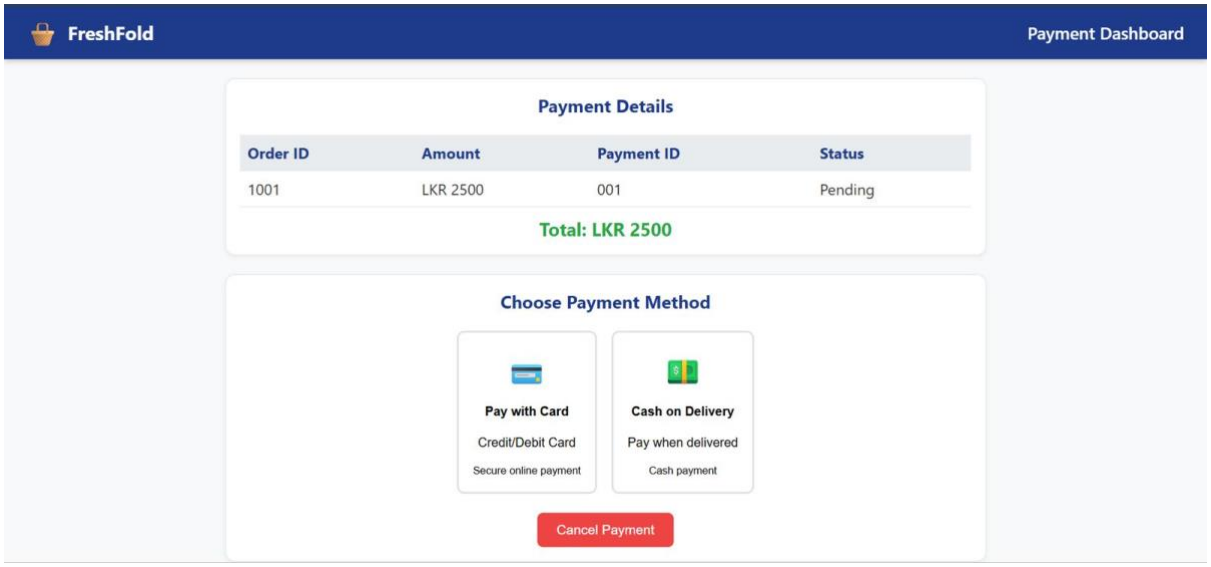

FreshFold Laundry
Home
Schedule Pickup
Track Order
Coordinator
Pickup Details
Pickup Date *
10/26/2025
Preferred Pickup Time *
9:00 AM
10:00 AM
11:00 AM
12:00 PM
1:00 PM
2:00 PM
3:00 PM
4:00 PM
5:00 PM
Delivery Details
Delivery Date *
10/27/2025
Preferred Delivery Time *
9:00 AM
10:00 AM
11:00 AM
12:00 PM
1:00 PM
2:00 PM
3:00 PM
4:00 PM
5:00 PM
Schedule Pickup & Delivery


FreshFold Laundry
Home
Schedule Pickup
Track Order
Coordinator
Driver
1 Unread Messages
4 Active Conversations
Active Messages & Conversations
Driver1
DRIVER_MESSAGE
10/21 04:13
REPLIED
c
Replies:
Driver1
10/21 04:13
d
Type your reply...
Reply
End Conversation
Driver1
ISSUE
10/21 04:42
REPLIED
delivered
Related to: Order #1
Replies:
Driver1
10/21 04:43
yes

4) Payment management







FreshFold

OTP Verification

Enter OTP

559502

Verify & Pay

FreshFold

Payment Receipt

Payment Successful ✅

Your payment has been verified.

Order ID: 1001

Status: Paid

Download Receipt

Redirecting to dashboard in 5 seconds...

FreshFold

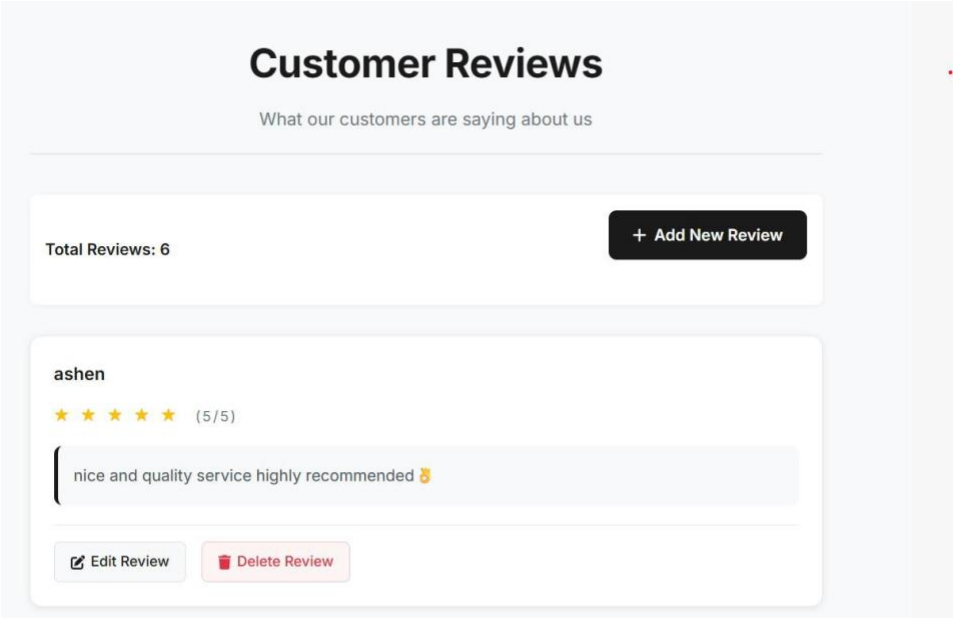
💵 Cash on Delivery

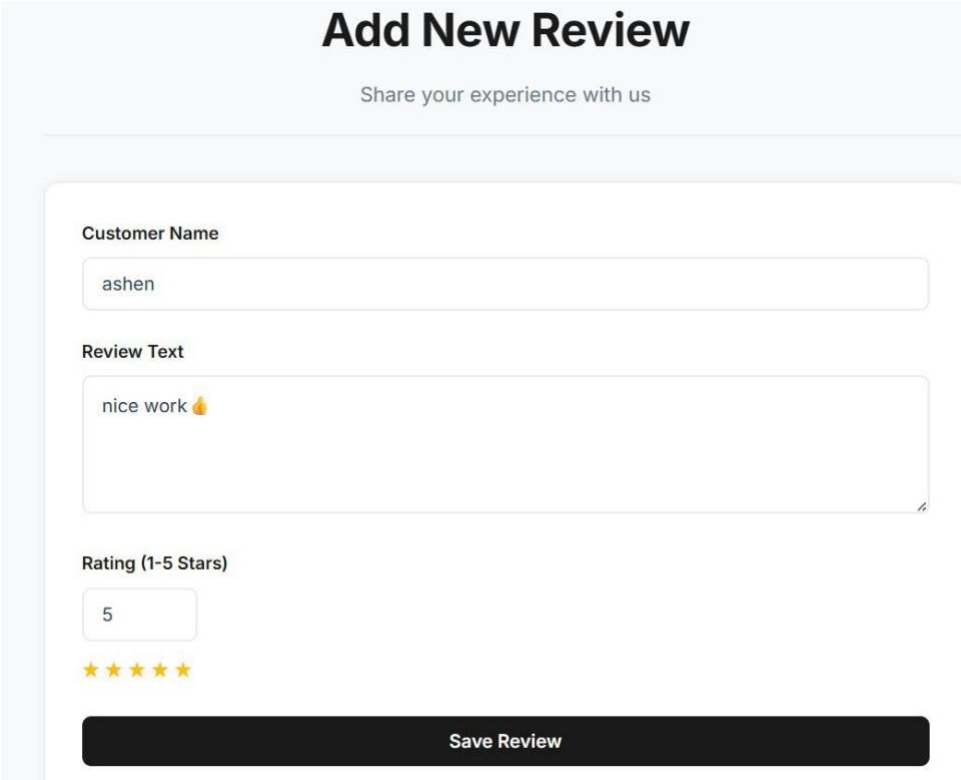
Amount to Pay: Rs 2500.00

You have selected **Cash on Delivery**.
Please confirm your order to proceed with payment upon delivery.

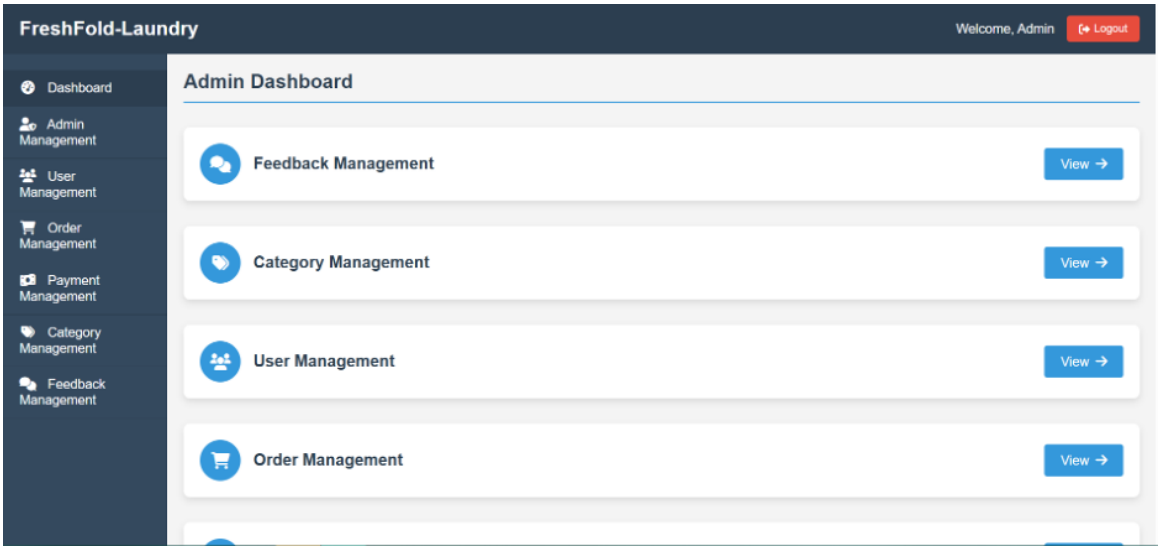
Confirm Cash on Delivery

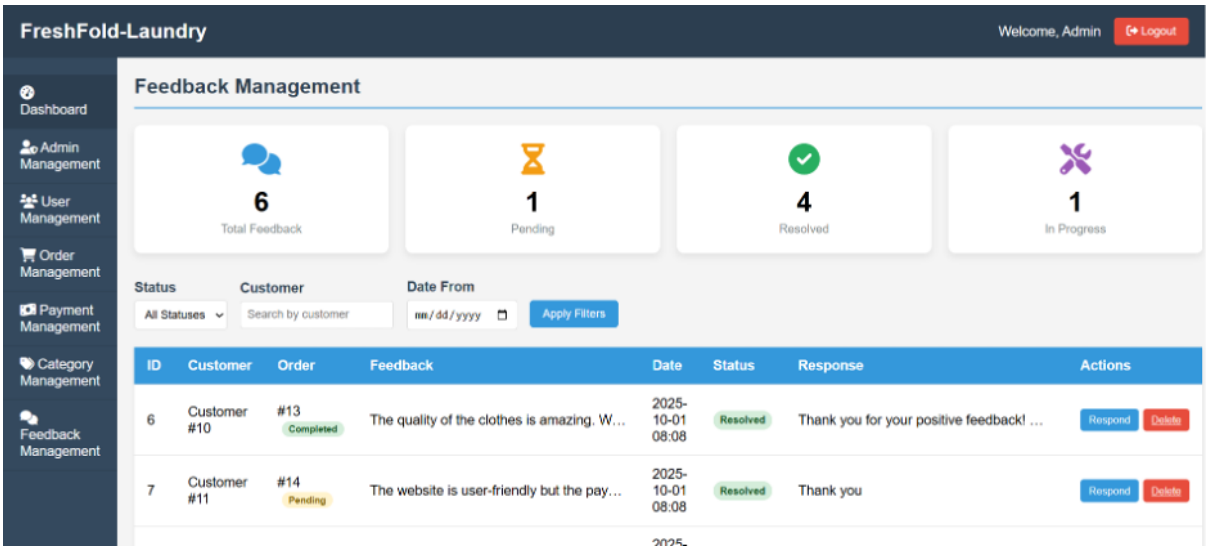
5) **Customer feedback and review management**

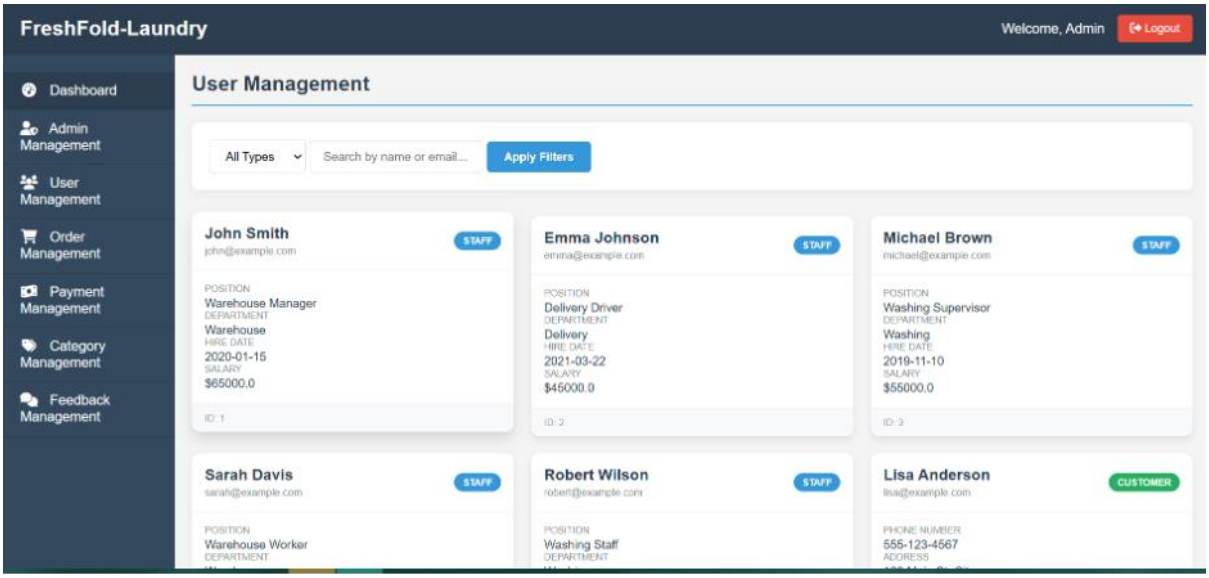


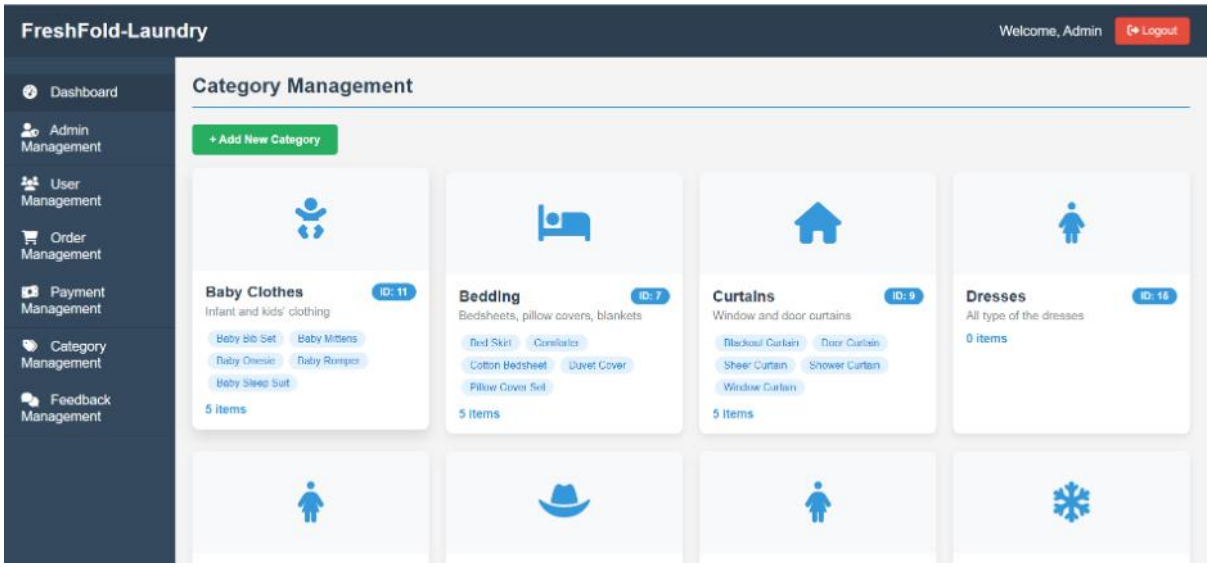


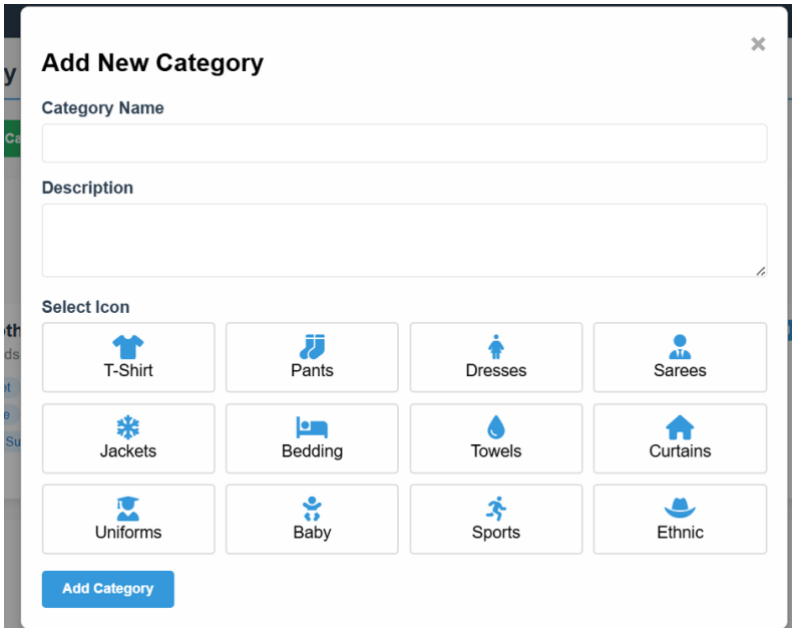
6) **Admin management**

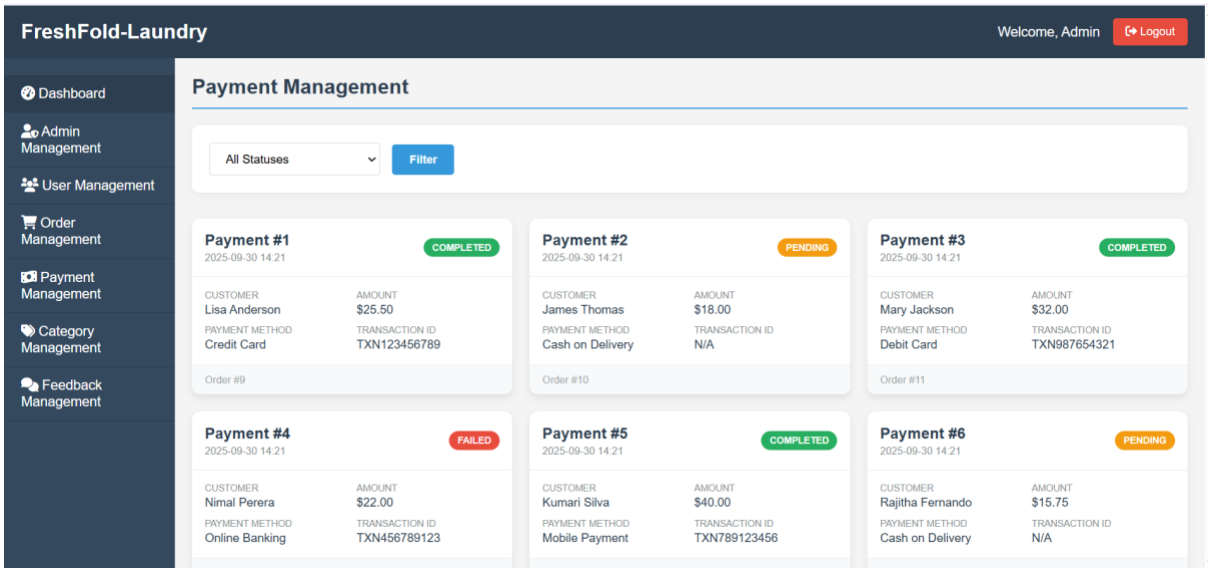







FreshFold-Laundry
Welcome, Admin
Logout
Dashboard
Admin Management
User Management
Order Management
Payment Management
Category Management
Feedback Management
Category Management
+ Add New Category
Baby Clothes
ID: 11
Infant and kids' clothing
Baby Bib Set
Baby Mittens
Baby Onesie
Baby Rompers
Baby Sleep Suit
5 items
Bedding
ID: 7
Bedsheets, pillow covers, blankets
Bed Skirt
Comforter
Cotton Bedsheet
Duvet Cover
Pillow Cover Set
5 items
Curtains
ID: 9
Window and door curtains
Blackout Curtain
Door Curtain
Sheer Curtain
Shower Curtain
Window Curtain
5 items
Dresses
ID: 16
All type of the dresses
0 items


Add New Category
Category Name
Description
Select Icon
T-Shirt
Pants
Dresses
Sarees
Jackets
Bedding
Towels
Curtains
Uniforms
Baby
Sports
Ethnic
Add Category


FreshFold-Laundry
Welcome, Admin
Logout
Dashboard
Admin Management
User Management
Order Management
Payment Management
Category Management
Feedback Management
Payment Management
All Statuses
Filter
Payment #1
2025-09-30 14:21
COMPLETED
CUSTOMER
Lisa Anderson
AMOUNT
$25.50
PAYMENT METHOD
Credit Card
TRANSACTION ID
TXN123456789
Order #9
Payment #2
2025-09-30 14:21
PENDING
CUSTOMER
James Thomas
AMOUNT
$18.00
PAYMENT METHOD
Cash on Delivery
TRANSACTION ID
N/A
Order #10
Payment #3
2025-09-30 14:21
COMPLETED
CUSTOMER
Mary Jackson
AMOUNT
$32.00
PAYMENT METHOD
Debit Card
TRANSACTION ID
TXN987654321
Order #11
Payment #4
2025-09-30 14:21
FAILED
CUSTOMER
Nimal Perera
AMOUNT
$22.00
PAYMENT METHOD
Online Banking
TRANSACTION ID
TXN456789123
Payment #5
2025-09-30 14:21
COMPLETED
CUSTOMER
Kumari Silva
AMOUNT
$40.00
PAYMENT METHOD
Mobile Payment
TRANSACTION ID
TXN789123456
Payment #6
2025-09-30 14:21
PENDING
CUSTOMER
Rajitha Fernando
AMOUNT
$15.75
PAYMENT METHOD
Cash on Delivery
TRANSACTION ID
N/A

5. Project Management

The laundry management system was developed using the Agile methodology. The tasks were divided into multiple sprints, each focusing on developing a specific functioning module. The use of Agile ensured consistent progress and transparency throughout the development of lifecycle.

Agile Approach and Sprint Summary

The development process was divided into 4 sprints using a SCRUM-based Agile approach.

- Sprint 1- Development of core functional modules including Order Management, Profile Management, Payment and billing, and task assignment functionality.
- Sprint 2- Manage user account access rights, implement order tracking, order assignment functionalities, Rating and Review module, and Profile updating functionality.
- Sprint 3- Implement Admin Dashboard functions and password reset functionality.
- Sprint 4- Implement Delivery and Pickup functionality, Conduct integration testing and system testing.

To ensure reliability and consistency with the objectives of the project, functional testing was conducted at the end of each sprint.

Task Distribution among Team Members

Module Assigned	**Responsible Team Member**
Order Management	Kodagoda D.S
Profile Management	Razeen M.R.M
Pickup and Delivery	Jayasinghe S.I.
Payment and Billing	Assalaarachchi B.T
Admin Dashboard	Asma M.F
Review & Complaints	Bandarayapa Y.A.R.J

Each member was responsible for both frontend and backend development of their assigned module, including UI design, API integration, validation logic, and testing.

Development Phases or Timeline

Week	Milestones
Week 3–4	Requirements gathering, role assignment, UML diagrams, system architecture
Week 5	Database design, backend setup, UI wireframes
Week 6–7	Sprint 1 & 2: Core module development, unit testing
Week 8–9	Sprint 3: Integration of modules
Week 10–12	Sprint 4: Final enhancements, UAT, system integration testing
Week 13–14	Documentation finalization, demo preparation, full system deployment

6. Conclusion and Future Work

Summary of Achievements

- A fully functional laundry management system was developed with core modules; order placement, profile management, delivery scheduling, payment processing, admin control, and review management.
- Secure login, role-based access, and password reset mechanisms were implemented.
- Card and COD payment flows were integrated with receipt of generation and payment status tracking.
- Real-time order status updates and task assignments for staff and drivers were enabled.
- A review and complaint system were designed with filtering and feedback response capabilities.

Challenges Faced

- Integrating independently developed modules resulted in API mismatches and inconsistencies in data flow. These were resolved using standardized DTO structures and collaborative debugging sessions.
- Expert collaboration was required to ensure consistent data validation between backend logic and frontend forms.
- Extensive role-based testing was necessary to validate functionality for administrators, employees, clients, and delivery coordinators.
- Cross-sprint parallel development occasionally resulted in overlapping logic and merging conflicts. Discipline in version control and frequent team meetings helped to resolve these.

Suggestions for Improvement or Extension

- Integrate SMS alerts for order updates and payment confirmations.
- Enable real-time chat between delivery coordinators and drivers.
- Add multi-language support for non-English users.
- Implement third-party payment gateways like Stripe or PayPal.
- Include map-based delivery tracking using Google Maps API.

7. Individual Contribution, Teamwork and Lessons Learned

Name	Specific role	Challenges faced	How you overcame those challenges	Key lessons learned	Reflection on the project
Razeen M.R.M	Developing Profile management function and handling role-based access to different sites	Challenges with Thymeleaf templates and encountered integration issues due to there were too many same classes in individual components instead of contributing to a shared main class used by the team.	Used unified main classes instead of separate component classes and maintained a single consistent framework.	This project taught the importance of early framework planning, agile sprint-based teamwork, and effective collaboration through shared version control to meet deadlines and avoid integration issues.	This experience highlighted the importance of thorough planning before implementation, It also emphasized the need for better team coordination through shared version control practices to ensure smoother collaboration and prevent integration issues.
Assalaarachchi B.T	Developing payment and billing function	Issues faced during integration with other components. Issues with the CRUD operations in the card payment function	Changed the structure of the code to match others and solved errors one by one.	Merging at least some components at first without waiting till last would have minimized integration errors.	This project gave me experience in managing a software development project in the real world effectively and working collaboratively with a team.
Asma M.F	Developing admin dashboard	Designing a user-friendly dashboard layout and handling UI errors.	Fixed layout issues through testing and improving code step by step.	Improved understanding of front-end-backend connection and UI design.	Learn to manage individual tasks efficiently and gain confidence in developing admin features.

Kodagoda D.S	Developing order management function	Getting errors when integrating with other components	Fixing errors collaboratively	Using git for version control early on, significantly reduces merging conflicts	Gained valuable experience in teamwork, effective time management, and collaborative project development
Bandarayapa Y.A.R.J	Developing feedback and review function	Creating a way for customers to add complaints via comments and ensuring staff could view them.	Implemented a simple form for reviews and a separate one for complaints, linking them to specific orders.	Learned how a feedback system directly supports the project's objective of "customer satisfaction through feedback and support."	Working on the review and complaint system showed me how software can bridge the gap between customers and management. It was rewarding to build a feature that enables the "Continuous improvement" mentioned in our major functions.
Jayasinghe S.I.	Developing Pickup & Delivery Scheduling System	Multi-role dashboard design, Time slot management and double-booking prevention.	First merge basic parts and tried to merge other components, Created separate dashboard templates with role-specific functionality, Fixed Using a function to check available time slots.	Early Integration Planning with team members, Agile sprint planning ensures manageable progress	This project significantly improved my full-stack development skills, particularly in Spring Boot and Thymeleaf integration. Also, I had a truly wonderful and collaborative teamwork experience.

8. Appendix

GitHub repository link

laundry management system